11.7.2 Objectives

- Understand how to perform SQL joins using Laravel's Query Builder.
- Use aggregate functions like `SUM`, `COUNT`, and `AVG` to summarize data.
- Apply filtering using `where()`, `whereBetween()`, and `whereNull()`.
- Display data in a Bootstrap-based view.

11.7.3 Prerequisites

- Laravel 12 installed
- MySQL or MariaDB database
- PHP 8.2+
- Composer
- Code editor (e.g., VS Code)

We will use **MySQL** for this lab, but you can adapt the code for SQLite or any other supported database.

> *This lab uses **PHP 8.4**, **Laravel 12** and **Visual Studio Code** as the primary editor.*

11.7.4 Steps

Here are the steps to create a simple report application using Laravel 12 Query Builder with joins and aggregate functions.

11.7.4.1 Step 1: Create a New Laravel Project

We will create a new Laravel project using the following command:

```
laravel new laravel12-querybuilder-joins
cd laravel12-querybuilder-joins
code .
```

Accept the default options for the installation including the database connection.

You should see the Laravel project structure in your code editor.

11.7.4.2 Step 2: Configure the `.env` File